Off-Page Signals Have Model-Specific Effects on Generative AI Search Visibility: Evidence from a Cross-Platform Audit of 2,729 Businesses Across Five Generative AI Systems

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Abstract

Background. Businesses are increasingly discovered through generative artificial intelligence (AI) search systems such as ChatGPT, Claude, Perplexity, Google Gemini, and Google AI Overviews. The off-page signals that drive whether a business appears in those answers are not well characterised, and prior work has typically aggregated outcomes across models, masking model-specific behaviour.

Methods. We audited 2,729 businesses across 14 verticals and four metropolitan markets (Los Angeles, New York, Chicago, Sydney). Each business was probed with approximately 95 intent-based prompts across each of five AI systems (ChatGPT GPT-4o, Claude Sonnet, Perplexity sonar-pro, Gemini 2.5-flash, and Google AI Overview via SerpApi), yielding 266,844 paired (business, model, prompt) observations. Each business was enriched with Domain Authority (Moz), organic-traffic and keyword metrics (SpyFu), on-page signals from a direct HTTP crawl (schema, FAQ, indexed pages, blog count, llms.txt), and twelve off-page presence indicators (Reddit, Quora, YouTube, LinkedIn, Wikipedia, Crunchbase, BBB, Yelp, Trustpilot, G2, Capterra, Google Business). We computed Pearson correlations (raw, log-transformed, and Spearman), partial correlations via OLS residualisation, an L2-regularised logistic regression with multicollinearity pruning by variance inflation factor, 95% bootstrap confidence intervals (5,000 resamples), and Benjamini–Hochberg false-discovery-rate (FDR) correction at $q = 0.05$.

Results. 69.5% of audited businesses (95% CI [67.8%, 71.2%]) were not mentioned by any of the five AI systems; only 1.5% (95% CI [1.06%, 1.98%]) were mentioned by all five. The most novel finding is model-specific: in per-model strict-isolation analyses (each model treated as a separate binary outcome, with nine controls including Domain Authority and SpyFu metrics), off-page signals including Reddit and Quora mention counts, Wikipedia presence, LinkedIn company pages, and BBB / Yelp / Trustpilot directory presence had small but robust positive partial correlations with mention probability in Claude (partial r up to +0.219), ChatGPT, and Perplexity, but were not significantly associated with mention probability in Gemini or Google AI Overviews. Of 60 model-by-predictor cells, 34 survived BH-FDR at $q < 0.05$. In paired within-response analyses, businesses whose URL was cited as a source in a Perplexity response were 6.03× more likely to also be mentioned in that response (95% CI [5.62, 6.46]); the corresponding lift for Google AI Overview was 6.64× (95% CI [2.48, 11.30]). Domain Authority, the headline predictor in earlier reporting, had a raw Pearson r of 0.337 but its multivariate logistic-regression coefficient confidence interval crossed zero (standardised $\beta = +0.139$, 95% CI [−0.031, +0.316]).

Conclusion. Off-page presence is statistically associated with AI mention probability for three of five major generative AI systems, with effect sizes that are small per individual signal (partial $r \approx 0.10$–0.22) but consistent across signals. The composite outcome used in earlier reporting averages over models that do and do not respond to off-page signals and consequently understates this relationship. A pre-registered controlled trial is planned to test causal magnitude. The dataset and analysis approach are released openly to enable replication.

Keywords

generative engine optimization; AI search; large language models; ChatGPT; Claude; Perplexity; Google AI Overviews; citation analysis; business visibility; search engine optimization; generative AI; LLM grounding

1. Introduction

Conventional web search has historically determined which businesses are seen by potential customers. Over the past three years that gatekeeping role has begun to shift to generative AI systems. ChatGPT, Claude, Perplexity, Google Gemini, and Google AI Overviews increasingly answer commercial-intent queries directly, often listing or recommending a small set of businesses without surfacing traditional search-engine results pages. The features that determine whether a given business appears in those AI-generated answers are poorly understood, and the practitioner discourse has moved faster than the empirical record.

This paper reports an observational audit of 2,729 businesses across 14 verticals and four metropolitan markets, with each business probed against five generative AI systems. The analysis asks three questions. First, which off-page signals (Reddit and Quora mentions, presence in structured directories such as Wikipedia, LinkedIn, BBB, Yelp, Trustpilot, and so on) are associated with whether a business is mentioned by an AI system, after controlling for site-level authority metrics (Moz Domain Authority, SpyFu domain strength, organic traffic) and on-page content signals? Second, do those associations differ systematically across AI systems, or do they generalise across ChatGPT, Claude, Perplexity, Gemini, and Google AI Overview? Third, when an AI system exposes the URLs it cites as sources, does citation as a source predict whether the business is mentioned in the response itself?

The contributions of this paper are three. (1) The dataset itself: 2,729 businesses, 266,844 paired (business, model, prompt) observations, and rich enrichment of authority, on-page, and off-page signals — released under a permissive licence. (2) A per-model analysis showing that several off-page signals have small but statistically robust associations with AI mention probability for three of five models tested, but not for the other two. (3) A pre-commitment to a controlled trial, described in §7, that will test causal magnitude. The author is the founder of MentionLayer, a commercial platform whose product motivation overlaps with the questions in this paper; this conflict of interest is disclosed in §3 and §12, and the data and analysis approach are released to enable independent verification.

This paper supersedes earlier reports of the same dataset. Version 1 (April 2026) reported headline statistics on a Phase 1 sub-sample of 1,004 businesses. Version 2 (May 2026, DOI 10.5281/zenodo.20048424) extended the analysis to a Phase 2 expansion but contained methodological issues including unreported skew in count features, multicollinearity in the logistic-regression design matrix, no multiple-comparison correction, and an aggregation of outcomes across models that masked the model-specific structure described in §4. The v2 record has been withdrawn. The differences between v2 and the present analysis are catalogued in a companion changelog.

2. Background and Related Work

Three strands of prior literature inform this study: retrieval-augmented generation and grounding in large language models; classical web-search ranking and authority signals; and the emerging practitioner literature on generative engine optimization (GEO).

2.1 Retrieval-augmented generation and source attribution

Lewis et al. (2020) introduced retrieval-augmented generation, demonstrating that conditioning a language model on retrieved documents improves factual accuracy on knowledge-intensive tasks.

Subsequent work has examined how language models attribute outputs to sources and whether those attributions are faithful. Liu, Zhang, and Liang (2023) evaluated source verification in commercial generative search engines and found that a substantial fraction of cited sources do not actually support the claims they accompany, motivating ongoing concern about grounding. Bohnet et al. (2022) proposed an attributable question-answering framework that has since been adopted in evaluation of AI search systems. Petroni et al. (2019) and Roberts, Raffel, and Shazeer (2020) probed the knowledge implicitly stored in large language model parameters, illustrating that some entity associations can be recalled without retrieval — a relevant background fact for interpreting why some AI systems may mention a business even when no source URL is exposed.

2.2 Search ranking, authority, and link-graph signals

The classical web-search literature established that link-graph signals such as PageRank (Page, Brin, Motwani, & Winograd, 1999) and authority measures derived from them predict ranking and discoverability. Modern proprietary metrics — including Moz Domain Authority — are descendants of this tradition. Wikipedia has been studied as both an information resource and a ranking factor; Mesgari et al. (2015) reviewed empirical work on Wikipedia's quality and influence, and McMahon, Johnson, and Hecht (2017) showed that Wikipedia content materially affects downstream search outcomes. Reviews and aggregate consumer signals have been studied as drivers of business discoverability and revenue (e.g., Luca, 2016, on Yelp ratings).

2.3 Reddit, Quora, and forum content as training and retrieval sources

Reddit and Quora are over-represented in publicly disclosed pre-training corpora (Gao et al., 2020; Penedo et al., 2023), which has motivated a practitioner thesis that mentions on these platforms may influence model outputs. The volume of Reddit-derived text in training data, together with the use of Reddit and Quora as retrieval targets in some commercial systems, makes off-page mentions on those platforms a plausible — though not previously well-quantified — driver of AI visibility.

2.4 Generative engine optimization and AI search behaviour

A growing practitioner literature treats generative AI systems as a new search surface and proposes interventions analogous to traditional search engine optimization. Aggarwal, Murahari, and Rajpurohit (2023) presented an early framework for generative engine optimization, examining how content structure influences inclusion in generative answers. Industry studies have documented disparities in business representation across AI systems, but these have generally relied on aggregated outcomes and small samples, and have not, to our knowledge, reported per-model partial-correlation analyses with multiple-comparison correction. The present study seeks to fill that gap.

3. Methods

3.1 Conflict of interest disclosure

The author is the founder and operator of MentionLayer, a generative engine optimization platform whose product thesis overlaps with the research questions of this paper. To mitigate this conflict, the full anonymised dataset and the analysis approach are released openly under CC-BY-4.0 to enable independent verification, and a pre-registered controlled trial is committed (see §7) with explicit advance commitment to publish null and negative results.

3.2 Sample

The audit covers 2,729 businesses across 14 verticals (personal injury law; real estate; dental; accounting; med spa; home services / plumbing; financial advisors; SaaS — CRM; SaaS — project management; digital marketing agencies; e-commerce — baby/kids/family DTC brands; boutique hospitality; insurance brokers and comparison platforms; personal-finance and money-management apps) and four metropolitan markets (Los Angeles, New York, Chicago, Sydney), arranged as 32 industry-city slots.

Sampling proceeded in two waves. Phase 1 (n = 1,004) sampled businesses URL-first, beginning with an industry-city directory query and verifying first-party domain ownership. Phase 2 (n = 1,725) sampled name-first, beginning with a business name and resolving the URL post-hoc via Apify Google SERP. The URL-resolution rate in Phase 2 was approximately 75%; 426 Phase 2 businesses had no detectable web presence and are therefore excluded from any analysis that conditions on URL (Domain Authority, SpyFu metrics, on-page crawl). Some Phase 2 URLs resolved to aggregator or directory pages rather than first-party sites; these are flagged in the dataset.

3.3 AI systems probed

Each business was probed against five AI systems: ChatGPT (OpenAI GPT-4o), Claude (Anthropic Claude Sonnet), Perplexity (Perplexity sonar-pro), Gemini (Google Gemini 2.5-flash), and Google AI Overview (via SerpApi's structured AI-Overview field on Google search results). For each (business, model) pair, approximately 95 prompts were issued across six prompt categories: direct recommendation, comparison, specific need, conversational, authority-seeking, and decision-oriented. Prompts were industry-specific and intent-driven — for example, "What are the best personal injury lawyers in Los Angeles?" — and constructed without reference to any specific business name. The total (business, model, prompt) observation count was 266,844. Probing was conducted at temperature 0 or the API default for systems that do not expose a temperature parameter, and was completed during April–May 2026.

3.4 Mention detection

For each AI response, we determined whether the queried business was mentioned. Detection used a two-stage matcher: (1) a heuristic stage that searched for the canonical business name, common aliases, and the business domain in the response text, with normalisation for case, punctuation, and whitespace; and (2) a Claude Sonnet–based verifier (temperature 0) that adjudicated borderline matches and rejected obvious common-word collisions (for example, the word "Apple" appearing in an unrelated context). On a 200-row hand audit the false-positive rate of the combined matcher was below 2%; the false-negative rate was not formally measured and is acknowledged as a limitation.

3.5 Visibility score

A composite visibility score in [0, 100] was computed per business as a weighted sum: 0.50 × share of model (proportion of all (model, prompt) tests in which the business was mentioned, scaled to 100); 0.25 × recommendation rate (proportion of mentions where the business was recommended, not merely listed); 0.15 × average prominence (a 0–100 prominence score adjudicated by the Claude Sonnet analyser, with 0 = absent, 50 = listed among many, 100 = top recommendation); and 0.10 × model coverage (number of distinct models mentioning the business, divided by 5, scaled to 100). The full implementation is in the released codebase. Per-model binary outcomes (mentioned_in_<model>) are derived directly and never depend on the composite score.

3.6 Predictor enrichment

Each business was enriched with three classes of predictors. Authority and traffic metrics were obtained via the Moz Links API (Domain Authority) and the SpyFu API (domain strength, monthly organic clicks, total organic keywords, organic growth rate, branded-keyword volume). On-page signals were obtained via direct HTTP crawl of the resolved domain (presence of schema markup, schema completeness score, citability heuristic, FAQ content, sitemap presence, blog post count, total indexed pages, presence of llms.txt and llms-full.txt, AI-crawler robots policy). Off-page presence on twelve platforms was determined via Apify Google SERP queries of the form site:<platform> "<business name>" with light disambiguation: Reddit (mention count, unique subreddits), Quora (mention count), YouTube (mention count, channel presence), LinkedIn company page, Wikipedia, Crunchbase, BBB, Yelp, Trustpilot, G2, Capterra, and Google Business Profile.

3.7 Sample sizes per analysis

Different analyses operate on different non-null subsets. The breakdown is given in Table 1.

Analysis	Effective n	Why excluded rows drop
Headline percentages (composite outcome)	2,729	None
Per-model mention rates	2,729	None
Raw correlations on off-page signals	2,726	Trivial nulls in boolean directory columns
Raw correlations involving Domain Authority	≈ 2,200	URL-unresolved businesses (n = 426) plus rows where Moz did not return DA
Raw correlations involving SpyFu metrics	≈ 2,295	URL-unresolved or SpyFu non-coverage
Logistic regression (multicollinearity-cleaned)	2,729	Mean-imputation for sparse features; no row exclusion
Strict isolation, composite outcome (full controls)	1,693	Restricted to rows with non-null DA, Moz, SpyFu, and all off-page features
Per-model strict isolation	1,693–1,695	Same as above; minor variation by predictor null pattern
Citation correlation, paired within-response	66,920 (Perplexity) / 29,305 (AIO)	Only responses where the model returned a non-empty sources_cited array

Table 1. Effective sample sizes per analysis. The strict-isolation analytic n of 1,693 is a non-random sub-sample (it requires Moz, SpyFu, and full off-page coverage) and findings on it should be interpreted accordingly.

3.8 Statistical methods

Pearson product-moment correlations were computed in three forms — raw, log-transformed (log(x+1) of any right-skewed count predictor), and Spearman rank — to expose any structure obscured by skew. Partial correlations were computed via OLS residualisation: for each (predictor X, control set Z) pair, X was regressed on Z and the visibility outcome Y was regressed on Z, and the Pearson correlation of the residuals was reported. Logistic regression used L2 regularisation ($\lambda = 0.5$) with features standardised to zero mean and unit variance; three a-priori-redundant aggregate features (review_platform_count, directory_count, off_page_composite_score) were dropped before fitting because each is a deterministic sum of individual binary flags. Variance inflation factors were computed on the standardised design matrix and features with VIF > 10 would have been pruned iteratively (in practice none required pruning; the maximum observed VIF was 7.16). Bootstrap 95% confidence intervals were computed by non-parametric percentile resampling (5,000 resamples) at the business level for headline percentages and correlations and at the paired-row level for citation lifts. Multiple-comparison correction used Benjamini–Hochberg false-discovery-rate adjustment at $q = 0.05$ across the family of correlation tests, and separately across the family of 60 per-model strict-isolation cells.

3.9 Data and code availability

The full anonymised dataset, all output JSON files, and the analysis scripts will be deposited on Zenodo with a separate DOI on or around the publication date of this preprint. The Phase 3 controlled-trial protocol described in §7 will be pre-registered on the Open Science Framework before the trial begins.

4. Results

4.1 AI mention is rare; cross-model agreement is rarer

Across the full sample of 2,729 businesses, 69.5% (95% CI [67.8%, 71.2%]) were not mentioned by any of the five AI systems tested. 30.5% (95% CI [28.8%, 32.2%]) were mentioned by at least one model, and only 1.5% (95% CI [1.06%, 1.98%]) were mentioned by all five. The mean number of models mentioning a business was 0.6. Per-model mention rates over the full set of probes were 6.1% for Perplexity, 4.2% for Google AI Overview, 3.9% for ChatGPT, 3.2% for Claude, and 2.8% for Gemini.

4.2 Per-model strict isolation is the headline finding

The most novel and robust finding in this study is that several off-page signals have small but statistically robust positive partial correlations with mention probability for ChatGPT, Claude, and Perplexity, but show no significant association with mention probability for Gemini or Google AI Overview. The analysis treats each model as a separate binary outcome (mentioned_in_<model>), uses nine controls (Domain Authority, Google review count, blog post count, schema score, citability score, SpyFu domain strength, monthly organic clicks, total organic keywords, organic growth rate), and applies BH-FDR across the full family of 60 (model × predictor) cells at $q = 0.05$. Of the 60 cells, 34 survive correction. Tables 2–4 give the headline rows.

Table 2. Reddit mention count → AI visibility, partial correlation by model.

Model	Partial r	95% CI	BH q-value	Significant
Claude	+0.190	[+0.147, +0.233]	5.9e-14	Yes
ChatGPT	+0.128	[+0.081, +0.174]	4.5e-07	Yes
Perplexity	+0.110	[+0.063, +0.156]	1.4e-05	Yes
Gemini	+0.031	[−0.015, +0.077]	0.305	No
Google AI Overview	+0.011	[−0.038, +0.061]	0.738	No

Table 3. Quora mention count → AI visibility, partial correlation by model.

Model	Partial r	95% CI	BH q-value	Significant
Claude	+0.203	[+0.155, +0.248]	1.0e-15	Yes
ChatGPT	+0.113	[+0.062, +0.163]	8.4e-06	Yes
Perplexity	+0.087	[+0.041, +0.137]	6.8e-04	Yes
Gemini	+0.025	[−0.020, +0.073]	0.416	No
Google AI Overview	+0.016	[−0.032, +0.066]	0.631	No

Table 4. Wikipedia presence → AI visibility, partial correlation by model.

Model	Partial r	95% CI	BH q-value	Significant
Claude	+0.219	[+0.172, +0.266]	4.5e-18	Yes
Perplexity	+0.124	[+0.077, +0.173]	8.9e-07	Yes
ChatGPT	+0.103	[+0.054, +0.152]	5.2e-05	Yes
Gemini	+0.005	[−0.039, +0.049]	0.839	No
Google AI Overview	−0.022	[−0.068, +0.027]	0.508	No

The same pattern holds for has_youtube_channel, has_linkedin_company, has_bbb, has_yelp, has_trustpilot, reddit_unique_subreddits, has_g2, has_capterra, and has_crunchbase: positive and BH-FDR-significant for Claude, ChatGPT, and Perplexity, with point estimates clustered between +0.07 and +0.22; null or near-null and not BH-FDR-significant for Gemini (with a single exception, has_yelp at +0.070, q = 0.008) and for Google AI Overview (with a single exception, has_bbb at +0.054, q = 0.045). The single largest off-page partial r anywhere in the table is has_wikipedia → Claude (+0.219), and the second largest is quora_mention_count → Claude (+0.203).

Two consequences follow. First, off-page presence is associated with AI mention probability for three of the five major generative AI systems studied here, after controlling for site authority, SpyFu traffic metrics, and on-page content signals. Second, any analysis that aggregates the binary outcome across all five models will under-state these effects, because the average dilutes models that respond to off-page signals against models that do not. This is the mechanism by which the v2 "strict-isolation" composite-outcome analysis appeared to find that off-page signals collapsed near zero (see §4.5).

4.3 Cited sources predict mention in Perplexity and Google AI Overview

Perplexity and Google AI Overview expose the URLs they cite as sources in their structured responses. ChatGPT, Claude, and Gemini do not (their sources_cited arrays are empty by API construction, not by absence of citations). The analysis below is therefore scoped to Perplexity and Google AI Overview only and must not be generalised to other systems.

Within the paired set of (response, business) rows where Perplexity returned a non-empty sources_cited array (n = 66,920 paired rows), a business whose URL appeared in sources_cited was 6.03× more likely to be mentioned in the response than a business whose URL did not appear (95% CI [5.62, 6.46]; phi = 0.206; P(mention | linked) = 0.339, P(mention | unlinked) = 0.051). For Google AI Overview (n = 29,305 paired rows), the corresponding lift was 6.64× (95% CI [2.48, 11.30]; phi = 0.050; P(mention | linked) = 0.314, P(mention | unlinked) = 0.042). At the per-business level, the count of cited-as-source events correlates with the composite visibility score for these two models (Perplexity: r = 0.228, n = 2,729; Google AI Overview: r = 0.106, n = 2,729) and is precisely zero for the three models that do not expose sources, by construction.

Categorising the 4,737 unique citation URLs observed across Perplexity and Google AI Overview, blog and other content sites accounted for the largest share (2,331 citations, 49.2%), followed by the businesses' own websites and aggregator profiles (1,971 citations, 41.6%), YouTube (186, 3.9%), industry directories (121, 2.6%), and news media (116, 2.5%). Reddit, LinkedIn, Yelp, Trustpilot, and Capterra accounted for fewer than ten cited URLs each in the source-URL stream from these two models — a finding that contrasts with the per-model strict-isolation result that those same platforms predict mention probability in Claude and ChatGPT, where source URLs are not exposed.

4.4 Domain Authority is associated but loses multivariate significance

Domain Authority has a raw Pearson correlation of 0.337 with the composite visibility score on the Phase 1 sub-sample (n = 1,004) and 0.217 on the full sample (n = 2,200 businesses with non-null DA). After BH-FDR correction it remains significant as a raw correlation. In the multivariate L2-regularised logistic regression on n = 2,729 — which simultaneously estimates 22 features — its standardised coefficient is +0.139 with a bootstrap 95% confidence interval of [−0.031, +0.316], which crosses zero. In other words, once the off-page presence indicators are admitted as joint predictors, the marginal contribution of Domain Authority is no longer distinguishable from zero in this sample. We interpret this as confounding rather than absence of effect: high-DA businesses are systematically more likely to have Wikipedia entries, LinkedIn company pages, BBB and Yelp listings, and active YouTube channels, and the off-page indicators absorb most of the explanatory power that DA appeared to carry alone. Table 5 lists the eight features whose logistic-regression bootstrap CI excludes zero.

Table 5. Logistic regression (n = 2,729): features with bootstrap 95% CI excluding zero, sorted by |coefficient|.

Feature	Std. coefficient	95% CI
spyfu_total_organic_keywords	−0.347	[−0.517, −0.126]
has_youtube_channel	+0.287	[+0.095, +0.460]
has_yelp	+0.222	[+0.093, +0.359]
has_bbb	+0.221	[+0.100, +0.342]
spyfu_domain_strength	+0.199	[+0.022, +0.375]
has_linkedin_company	+0.168	[+0.006, +0.330]
citability_score	+0.123	[+0.008, +0.244]
has_google_business	+0.094	[+0.002, +0.188]

Train accuracy on the L2 logistic model was 73.3% against a 30.5% positive base rate. The negative coefficient on spyfu_total_organic_keywords is interpreted in the presence of spyfu_monthly_organic_clicks and spyfu_domain_strength as a residualised effect rather than a marginal effect of organic keyword count, and is reported here for completeness rather than as a substantive finding.

4.5 Composite-outcome strict isolation: a supporting analysis

An earlier version of this analysis (v2) reported that, in a strict-isolation specification using the composite visibility score as the outcome and 22 controls (Domain Authority, Google reviews, blog count, schema score, citability, SpyFu domain strength, monthly organic clicks, total organic keywords, organic growth rate, plus the off-page features themselves), the partial correlations of off-page signals with visibility collapsed close to zero. We reproduce that result on n = 1,693 in Table 6 — but the present paper treats it as a supporting analysis rather than a headline.

Table 6. Strict isolation, composite outcome (22 controls, n = 1,693): top features by |partial r| with bootstrap 95% CI.

Feature	Partial r	95% CI	Significant
press_mention_count_12mo	−0.093	[−0.137, −0.047]	Yes
has_capterra	−0.088	[−0.145, −0.030]	Yes
has_yelp	+0.074	[+0.033, +0.115]	Yes
has_bbb	+0.063	[+0.012, +0.111]	Yes
has_linkedin_company	+0.043	[−0.003, +0.088]	No

The reason the composite outcome appears to attenuate effects is structural rather than empirical. The composite score averages mention probability across all five AI systems. Per the per-model results in §4.2, off-page signals predict mention probability in three of those five and are null for the other two. Averaging dilutes a real effect of partial r ≈ +0.10 to +0.22 in three models with effects near zero in two models. The composite analysis therefore obscures rather than refutes the per-model finding, and the reader should treat §4.2 as the load-bearing result.

4.6 Skew correction: log-Pearson and Spearman

Several count predictors are right-skewed. Pearson correlations on the raw scale are influenced more by long-tail outliers than by the bulk of the distribution. We therefore re-computed Pearson correlations on log(x+1)-transformed predictors and rank correlations (Spearman ρ) on the same predictors. Table 7 lists the features whose relationship changes materially under transformation.

Table 7. Skew sensitivity: raw Pearson, log-Pearson, and Spearman ρ for selected count features.

Feature	Raw r	Log r	Spearman ρ	n	Note
moz_total_external_links	−0.059	+0.099	+0.107	1,200	Sign flips under transform
moz_root_domains_to_root_domain	−0.068	+0.118	+0.122	1,200	Sign flips under transform
total_indexed_pages	−0.014	+0.141	+0.133	2,300	Null → real positive
google_review_count	+0.154	+0.307	+0.278	2,554	Magnitude approximately doubles
spyfu_total_organic_keywords	+0.002	+0.168	+0.189	2,295	Null → real positive
spyfu_monthly_organic_clicks	+0.047	+0.189	+0.202	2,295	Substantially stronger
reddit_mention_count	+0.337	+0.337	+0.330	2,726	Robust either way
quora_mention_count	+0.334	+0.335	+0.320	2,726	Robust either way
youtube_mention_count	+0.355	+0.357	+0.339	2,729	Robust either way

The headline forum-mention correlations (Reddit, Quora, YouTube) are insensitive to the choice of Pearson on raw counts, Pearson on log counts, or Spearman rank — they are stable on all three. By contrast, several Moz backlink and SpyFu traffic metrics had wrong-sign or attenuated raw Pearson correlations driven by skew; under log transform they take their expected positive sign with small-to-moderate magnitudes.

4.7 Directory-presence lift by Domain Authority quartile

We examined how the visibility uplift associated with directory presence varies with site authority. We split the sample into Domain Authority quartiles and compared mean visibility scores between businesses with above-median directory_count and below-median directory_count within each quartile.

Table 8. Directory-presence lift on visibility score by DA quartile (n ≈ 550 per quartile).

DA bucket	Avg vis (low directory)	Avg vis (high directory)	Δ
Q1 (DA 1–20)	2.4	7.2	+4.8
Q2 (DA 20–39)	9.6	14.1	+4.5
Q3 (DA 39–59)	11.1	19.5	+8.5
Q4 (DA 59–98)	7.0	27.2	+20.2

The directory-presence uplift is small at low Domain Authority (≈ +4.8 visibility points in Q1) and large at high Domain Authority (+20.2 visibility points in Q4). One reading is that off-page presence has the most leverage for already-authoritative businesses; another is that directory coverage itself becomes more complete as DA rises, and the uplift in Q4 reflects the joint product of authority and coverage rather than a pure off-page effect. The dataset is consistent with either reading; the controlled trial in §7 is designed to discriminate.

4.8 Industry variation

Per-industry invisibility rates (composite outcome) range from 28% in personal-injury law (Los Angeles) to 94% in home services (New York). Per-industry top predictors are broadly consistent with the pooled analysis, with off-page presence indicators and directory counts ranking among the top five predictors in 27 of 32 industry-city slots. We do not list the full per-industry breakdown in the paper for space; it is available in the released dataset.

5. Discussion

The most important finding of this study is that off-page signals do not act uniformly on all generative AI systems. Reddit and Quora mention counts, Wikipedia presence, LinkedIn company pages, and several directory listings have small but robust positive partial correlations with mention probability in Claude, ChatGPT, and Perplexity. The same predictors are not statistically associated with mention probability in Gemini or Google AI Overview after multiple-comparison correction. This pattern is novel — to the author's knowledge no published audit has previously reported per-model partial-correlation analyses on a sample of this size with FDR correction — and it has consequences both for empirical claims about "AI visibility drivers" in the abstract and for practitioner strategy.

Why might Gemini and Google AI Overview behave differently from Claude, ChatGPT, and Perplexity? Several non-exclusive explanations are consistent with the data. Gemini and Google AI Overview are both products of Google; their training-data composition, retrieval pipeline, and grounding behaviour are determined by a single organisation. Both may rely more heavily on Google's own knowledge graph, on first-party Google Business Profile data, and on classical search-ranked documents than on the Reddit and Quora content that has been observed in the open pre-training corpora of other models. The source-URL analysis in §4.3 is consistent with this: Google AI Overview's cited sources, where exposed, are dominated by businesses' own websites, blog content, and YouTube — not by the off-page platforms that show partial-correlation effects in Claude and ChatGPT.

A second implication is for the practitioner literature. The recommendation, common in industry discourse, that businesses pursue "shotgun" off-page presence to improve AI visibility is neither uniformly supported nor uniformly refuted by these data. It is supported for three of the five major systems studied here, with effects per individual signal that are small (partial $r \approx +0.10$ to $+0.22$) and additive across signals. It is not supported as a strategy for being mentioned in Gemini or Google AI Overview, where directory presence and forum mentions appear to carry no marginal weight in this sample. Effective strategy therefore depends on which AI system a business prioritises.

A third implication concerns Domain Authority. The raw correlation of DA with the composite visibility score is 0.337 on the Phase 1 sub-sample and 0.217 on the full sample — a moderate association. In the multivariate logistic regression that includes off-page presence indicators as joint predictors, DA's coefficient confidence interval crosses zero. We interpret this not as absence of effect but as confounding: high-DA businesses are systematically more likely to have Wikipedia entries, LinkedIn company pages, and structured directory presence, and the off-page indicators absorb most of the explanatory power that DA appeared to carry alone. Earlier reports that placed DA at the top of the predictor table were not wrong about its raw association; they were under-specifying the multivariate structure.

All of these findings are correlational. The dataset cannot adjudicate causality. A business that is mentioned on Reddit and on Wikipedia and that ranks in BBB and Yelp directories is likely to be a more established business with more brand equity, and brand equity itself plausibly drives both off-page presence and AI mention probability. The controlled trial described in §7 is designed to measure causal magnitude.

6. Limitations

Observational, not causal. The dataset cannot distinguish causation from confounding. Partial-correlation and OLS-residualisation analyses isolate linear contributions of a predictor controlling for a measured set of confounders; unobserved confounders (brand age, marketing spend, executive PR effort, demand-side prominence) are not controlled for. Phase 3 (May–July 2026) is a controlled trial with random assignment to mitigate this.

URL-dependent signals operate on $n \approx 2{,}303$. 426 of 2,729 Phase 2 businesses (15.6%) had no detectable web presence. They are excluded from any analysis that requires a resolved domain — Domain Authority, SpyFu metrics, on-page crawl, source-URL categorisation. They are still included in raw correlations on off-page presence and in the logistic regression with mean-imputation.

Source-URL analysis covers two of five models. ChatGPT, Claude, and Gemini do not expose sources_cited via their public APIs. The §4.3 finding (6.03× lift in Perplexity, 6.64× lift in Google AI Overview when a business URL is cited as a source) speaks only to those two systems. It must not be generalised to ChatGPT, Claude, or Gemini.

Phase 2 sampling and URL resolution. Phase 2 businesses were sampled by name within geo-stratified industry slots and resolved to URLs post-hoc via SERP. A subset of resolved URLs are aggregator or directory pages rather than first-party sites; this is flagged in the dataset. Selection bias favouring geographically findable businesses is possible.

Stochastic AI outputs. Each (business, model, prompt) cell is a single sample from a stochastic process. We did not measure within-cell run-to-run variance. Multiple prompts (≈ 95 per business per model) provide some averaging across prompts but not across runs of the same prompt.

Strict-isolation analytic n is non-random. The 1,693-row strict-isolation subset requires non-null Domain Authority, full SpyFu coverage, and complete off-page features. This sub-sample over-represents larger and better-indexed businesses; effect sizes within it may not generalise to the long tail of the small-business population.

Single point in time. All probing was conducted in April–May 2026. AI model versions, training cutoffs, and grounding behaviour change frequently. Replication on the same sample at a later date would likely produce a different mention pattern at the margin.

Vertical and market sample skew. 14 verticals × 4 metropolitan markets are broad but not representative of the global business population. Effects measured in legal, dental, SaaS, and professional-services contexts are not guaranteed to generalise to heavy industry, government, consumer goods at scale, or non-English markets.

Brand-matching error. The combined heuristic and Sonnet-verified matcher had a hand-audited false-positive rate below 2% on a 200-row sample. The false-negative rate was not formally measured. Common-word brand names are at higher risk of either error type.

Conflict of interest. The author founded MentionLayer, a commercial GEO platform, and the research questions overlap with the platform's product thesis. Mitigations are open data release and pre-registered Phase 3.

7. Conclusion and Future Work

Across an audit of 2,729 businesses, 14 verticals, four metropolitan markets, and 266,844 paired (business, model, prompt) observations against five major generative AI systems, this study finds: (1) two-thirds of the audited businesses are not mentioned by any AI system, with wide bootstrap-supported confidence intervals around that headline; (2) several off-page signals — including Reddit and Quora mention counts, Wikipedia presence, LinkedIn company pages, and BBB / Yelp / Trustpilot directory listings — have small but robust positive partial correlations with mention probability in Claude, ChatGPT, and Perplexity, after controlling for site authority and on-page signals, and after FDR correction; (3) those same signals are not significantly associated with mention probability in Gemini or Google AI Overview; (4) within Perplexity and Google AI Overview, where the systems expose source URLs, citation as a source predicts being mentioned in the response with a 6× lift; and (5) Domain Authority's marginal effect after controlling for off-page presence is not distinguishable from zero in this sample, consistent with strong confounding between DA and structured off-page presence.

These results are correlational. To test causal magnitude, a Phase 3 controlled trial is planned for May–July 2026, with the protocol pre-registered on the Open Science Framework before enrolment. The intended design recruits 25–30 businesses with low baseline AI visibility, applies a structured off-page seeding intervention (planted Reddit, Quora, LinkedIn, and directory presence) over a 30–45 day window, and measures change in per-model mention probability against a held-out within-business preperiod and against a non-treated comparison set. Six pre-specified success thresholds will be evaluated; the trial will be considered successful if at least four of six are met. The author commits in advance to publishing the results regardless of direction, including null and negative results.

Data and Code Availability

The full anonymised dataset (2,729 businesses, all enrichment fields, all per-model outcomes) and the analysis scripts will be deposited on Zenodo with a separate DOI on or near the publication date of this preprint. The Phase 3 protocol will be pre-registered on the Open Science Framework before the trial begins. Inquiries: joel@xpanddigital.io.

Conflict of Interest Declaration

The author is the founder and operator of MentionLayer, a generative engine optimization platform whose product thesis overlaps with the research questions in this paper. To mitigate this conflict, the full anonymised dataset and the analysis approach are released openly under CC-BY-4.0 to enable independent verification, and the Phase 3 controlled trial is pre-registered with an advance commitment to publish null and negative results.

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